Counting Fixed-Knot Completions in the Standard-Eleven Mosaic Model

Anonymous Authors

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Abstract

We study completion counting for the fixed eleven-tile Lomonaco–Kauffman knot-mosaic alphabet. An input is only a finite partial assignment of ordinary microscopic tiles; every hole admits all eleven tiles. We define a polynomially recognizable crossing-blocked exact-core class on which every legal one-component completion is automatically a planar unknot. By a solution-bijective reduction from maximum-degree-three grid Hamiltonicity with marked forced edges, we prove that counting these raw completions is `#P`-complete under parsimonious reductions. The proof is microscopic: optional corridors are Boolean, vertex cells select a unique degree-two mask, and completed mosaic components are exactly the cycles of a spanning two-factor.

For every fixed classical knot `J`, we then replace one forced corridor by a frozen two-terminal long-knot diagram of `J`. Every legal completion becomes the split union of `J` with planar unknots, without changing a single free choice. Counting completions of knot type `J` is therefore parsimoniously `#P`-complete in the raw unit-tile model. On the same syntactic family, Jones-polynomial equality with `V_J`, determinant equality with `det(J)`, and positivity of the Jones value at `-1` select exactly the one-component completions and are likewise parsimoniously complete. These are counting results over exponentially many completions; evaluating `V_L(-1)` for one diagram remains polynomial-time.

We also identify the exact unrestricted boundary. Exact-core raw unknot counting is `#P`-hard and belongs to `SpanP`; it belongs to `#P` if and only if mosaic unknot recognition belongs to `UP`. Counting distinct blank-padded rectangles is zero or infinite rather than a `#P` function. Finally, the guarded functions are fixed-parameter tractable by hole count and board width, while an FPRAS would imply `NP=RP`.

1 Introduction

Knot mosaics encode link diagrams by the eleven oriented local tiles introduced by Lomonaco and Kauffman. Existing work establishes equivalence with tame-knot theory after stabilization, develops mosaic and tile numbers, enumerates small mosaics, and implements SAT/ALLSAT and modern SageMath tools. The question here is different: how hard is it to count completions of a partially frozen microscopic mosaic when the completed diagram must have one specified topology?

There are two traps. First, a role-labelled macrotile reduction does not automatically prove a theorem about arbitrary unit-cell choices. Second, unknot recognition being in `NP` does not automatically put the number of distinct unknot completions in `#P`: one diagram may have many accepting topological certificates. Our reduction avoids both issues. Every hole is a genuine unrestricted standard-eleven cell, while a local blocked-side guard prevents all four-port tiles. Thus topology collapses to a deterministic component-count test on every input in the promised syntactic class.

The hardness source is the ASP-completeness theorem of the MIT Hardness Group et al. for undirected maximum-degree-three grid Hamiltonicity. ASP here is the solution-bijection notion, not answer-set programming. The constructive proof supplies marked edges forced in every output solution. Retaining those marks gives exactly the local forcing needed at a mosaic vertex while preserving the solution bijection.

The fixed-target extension is unusually clean. A fixed long-knot tangle is frozen into one already-forced corridor. Replacing it by a boundary-parallel arc preserves every component and every free tile choice. With the tag present, precisely one planar component is tied into `J`; all others remain split unknots. The result therefore covers every fixed classical knot and also gives genuine Jones/determinant selectors without invoking the open question whether the Jones polynomial detects the unknot.

1.1 Contributions

1. A parsimonious `#P`-completeness theorem for raw standard-eleven one-component unknot completion on a polynomially recognizable domain.
2. A literal microscopic bijection with forced-edge spanning two-factors, including polynomial area, malformed inputs, and multiplicity one.
3. For every fixed knot `J`, a raw tagged theorem for knot type, Jones equality, determinant equality, and determinant positivity.
4. The exact `SpanP/UP` boundary for unrestricted exact-core unknot counting.
5. A proof that literal all-rectangle counting is zero or infinite.
6. Exact fixed-parameter algorithms and an FPRAS barrier.

2 Related work

Lomonaco–Kauffman introduced the fixed eleven-tile model (2008); Kuriya–Shehab proved its stable equivalence with tame knots (2014). Mosaic-number and enumeration foundations include Lee–Hong–Lee–Oh (2014) and Oh–Hong–Lee–Lee (2015). Miller's SAT and ALLSAT work (2022, 2023) is the closest direct computational predecessor, but does not prove fixed-knot completion hardness. Heap et al. provide a current low-crossing tabulation (2025); Deng et al. provide the 2026 KnotMosaics SageMath package (arXiv:2605.14189).

Valiant introduced `#P` (1979). Köbler–Schöning–Torán introduced span functions, counting the distinct outputs of an NP transducer (1989). That is the appropriate upper bound when an object is guessed together with a potentially nonunique certificate. Unknot recognition is in `NP` by Hass–Lagarias–Pippenger (1999); Lackenby gives efficient knottedness certificates (2021). Neither result supplies certificate uniqueness.

The source theorem is FUN 2024, Theorem 18 (DOI 10.4230/LIPIcs.FUN.2024.23); the May 2026 full revision is arXiv:2405.08377. The latter updates authorship and puzzle applications; the proceedings result is the peer-reviewed hardness source.

For the Jones consequences, Kauffman's state model supplies normalization (198790009-7)), and Boninger gives a recent direct Jones/Goeritz treatment of `det(K)=|V_K(-1)|` (2023). Our result does not contradict the easy evaluation at `-1`: it counts completions satisfying that predicate.

No predecessor satisfying all of the following axes was located in the documented primary-source search through 14 August 2026: fixed standard-eleven library, raw partial-matrix input, exact one-component fixed-knot acceptance, and a parsimonious counting reduction. This is a corpus-scoped search result, not an absolute priority claim.

3 Model and finite counting convention

Let `T={0,...,10}` be the oriented standard-eleven alphabet. Tile `0` is blank; tiles `1,...,6` have one noncrossing arc; tiles `7,8` have two disjoint arcs; and `9,10` are the two crossings. Rotations are distinct tile IDs.

A seed is a canonical finite partial map

$$s : \mathbb{N}^2 \rightharpoonup T$$

with nonnegative, duplicate-free coordinates. A nonempty seed recovers

$$B_s = [0, h) \times [0, w), \quad h = 1 + \max\{r : (r, c) \in \operatorname{dom} s\}, \quad w = 1 + \max\{c : (r, c) \in \operatorname{dom} s\}.$$

A witness is one row-major tile value for every hole in `B_s`. Adjacent ports must match and no occupied port may leave the board. Strand components are cycles of the resulting degree-two graph, keeping the two strands at a crossing distinct.

A seed is **crossing-blocked** when:

1. it has no frozen crossing;
2. every hole has a side facing the exterior or a frozen blank neighbor;
3. `hw <= max(1,|dom(s)|)^2`;
4. every boundary cell is frozen blank; and
5. at least one interior cell is frozen nonblank.

Malformed and non-guarded inputs have value zero. This makes the counting function total and the board polynomial in its encoding.

3.1 Lemma — guarded normal form

Every suitably connected filling of a crossing-blocked seed is crossing-free. It is an unknot exactly when it has one strand component.

Proof. Every four-port tile occupies the blocked side, so tiles `7-10` are impossible at a hole; frozen crossings are forbidden. A closed connected one-manifold embedded in the plane is a simple closed curve. □

4 The parsimonious raw reduction

Define `#FGridHam_3(G,F)` to count Hamiltonian edge sets of an undirected maximum-degree-three grid graph `G` that contain the explicitly supplied set `F`, where `F` is incident to every vertex. Invalid instances have value zero. Retaining the marked black/tentacle edges in the constructive proof of FUN 2024, Theorem 18, and marking an arbitrary incident edge at any remaining degree-two vertex yields this total marked source without changing any solution. It is parsimoniously `#P`-complete.

Translate a connected source graph to nonnegative coordinates and scale the grid by four. Freeze every internal cell of an edge in `F` to its unique path tile. Leave every source vertex and every internal cell of an optional edge free. Freeze all remaining cells blank and add a one-cell blank moat. A connected `n`-vertex unit-grid graph has coordinate span at most `n-1`, so the compiled dense board has polynomial area even when source coordinates are binary.

4.1 Lemma — optional corridors

An optional corridor is either entirely blank or entirely its unique path tiles.

Proof. At an internal corridor cell both transverse neighbors are frozen blank. Exactly two standard tiles are compatible: blank and the unique tile joining predecessor to successor. Port matching propagates that Boolean choice along the whole corridor. □

4.2 Lemma — vertex selection

Every suitable filling selects exactly two source edges incident to each vertex and contains every edge in `F`.

Proof. A forced corridor occupies at least one side. Maximum degree three leaves a blank-blocked side, excluding four-port tiles. Every standard tile has even occupied degree, so the free vertex cell has degree exactly two; its two-side mask determines a unique tile. □

4.3 Lemma — microscopic bijection

Suitable fillings are in bijection with spanning two-factors containing `F`, and mosaic components are the cycles of the two-factor.

Proof. Lemmas 4.1–4.2 map a filling to a consistent selected edge set of degree two. Conversely, a two-factor uniquely sets every corridor to blank or its path tile and uniquely sets each vertex tile. The two maps are inverse, and strand tracing follows graph cycles. □

4.4 Theorem — raw parsimonious completeness

Let `#CB-RKW11_U(s)` count one-component unknot fillings of a crossing-blocked exact core. Then

$$\#\text{CB-RKW11}_U(s_{G,F}) = \#\text{FGridHam}_3(G, F),$$

and `#CB-RKW11_U` is `#P`-complete under parsimonious reductions.

Proof. By Lemma 4.3, one component is exactly one two-factor cycle, hence a Hamiltonian cycle. Lemma 3.1 makes it an unknot. A `#P` machine guesses one four-bit code per hole, rejects codes `11-15`, and checks ports and components deterministically. Each matrix has exactly one accepted code. □

5 Every fixed target knot

Fix an unoriented tame classical knot `J`. Choose a diagram, remove a noncrossing subarc adjacent to the unbounded face, rectilinearize the remaining one-string tangle on the standard-eleven grid, and add a blank collar. Denote the resulting two-terminal matrix by `T_J`. Let `T_0` be a boundary-parallel arc on the same rectangle with the same two gates. Both are constants because `J` is fixed.

A seed is **`J`-guarded** when it has one fully frozen translated copy of `T_J`, every frozen crossing lies in that copy, every hole has a frozen blank side, the area is quadratically bounded, and the outer boundary is a frozen blank moat. This is recognized by scanning constant-size translated patterns; no knot-equivalence test occurs in the guard.

Replace the tag by `T_0` without changing the holes. The resulting completed matrix is planar and has the same components. Therefore every suitable `J`-guarded completion with `k>=1` components has the complete topology

$$L(P) \cong J \sqcup U^{\sqcup(k-1)}. \tag{5.1}$$

Indeed, the tag ties J into the unique planar component passing through its two-point ball, while the other planar components remain split unknots.

To compile hardness, choose the first marked forced edge, rotate the entire grid orientation-preservingly if needed, enlarge the constant grid scale, and replace that forced corridor by T_J and two frozen leads. All optional cells and vertex cells are exactly those of the untagged reduction. Tag erasure gives a component-preserving bijection with the original fillings.

5.1 Theorem — raw fixed-target completeness

For every fixed classical knot J, counting J-guarded raw standard-eleven fillings of knot type J is #P-complete under parsimonious reductions. Hardness holds with all holes initially permitting all eleven tiles and with only the constant number of crossings in the frozen tag.

Proof. The source-to-filling bijection survives tag insertion. By (5.1), a filling has knot type J exactly when its two-factor has one cycle. For membership, guess each hole once and accept by deterministic local checks and component count; (5.1) supplies the semantic equivalence. □

5.2 Corollary — genuine Jones and determinant selectors

Let V_U(t)=1 and delta(t)=-(t^{}{1/2}+t^{}{-1/2}). On every suitable J-guarded completion,

$$V_{L(P)}(t) = \delta(t)^{k-1} V_J(t). \tag{5.2}$$

Consequently the following four predicates are equivalent on this family:

1. k=1;
2. L(P) has knot type J;
3. V_{L(P)}(t)=V_J(t);
4. |V_{L(P)}(-1)|=det(J); equivalently |V_{L(P)}(-1)|>0.

If k>1, (5.2) differs from V_J in the Laurent integral domain, and delta(-1)=0; if k=1, equality is immediate. Counting completions satisfying any predicate is therefore parsimoniously #P-complete. A Goeritz determinant computes the fourth predicate for one diagram in polynomial time. The theorem concerns counting all completing matrices.

6 Unrestricted counting and padding

Let F_raw(s) count one-component unknot fillings of a polynomially bounded exact core without the crossing-blocked syntax. It is #P-hard because the guarded instances are ordinary raw instances. It belongs unconditionally to SpanP: an NP transducer guesses a matrix and an unknot certificate, verifies them, and outputs only the matrix, so different certificates for the same matrix are collapsed.

Moreover,

$$F_{raw} \in \#P \quad \Longleftrightarrow \quad \mathrm{UNKNOT}_{mosaic} \in UP. \tag{6.1}$$

If F_raw has a #P machine, freeze every cell of a supplied closed mosaic; its count is zero or one, giving an unambiguous unknot machine. Conversely, combine a canonical filling guess with an unambiguous unknot verifier.

If each larger rectangle is a distinct witness, any accepting core extends by arbitrarily many blank rows and columns. The literal all-size count is therefore zero or infinite. The exact-core

convention is equivalent to a quotient by exterior blank padding on guarded instances: the moat prevents a strand crossing, and any nonblank exterior produces an additional component.

7 Algorithms and approximation boundary

With `h` holes, exact enumeration takes `O(11^h poly(|B_s|))` time. After erasing the fixed tag, a width-`w` scan stores the occupied frontier ports, their noncrossing connectivity, and whether a component has closed. There are Catalan-many states, giving `|B_s| 2^{O(w)}` arithmetic operations. Hardness already holds at crossing count zero for `J=U`, at the constant tag crossing count for fixed `J`, and at target component count one.

A parsimonious reduction transfers any FPRAS to `#SAT`. SAT self-reduction, amplification, and final witness verification then give an RP algorithm for SAT. Thus an FPRAS for any guarded fixed-target count implies `NP=RP`; a deterministic multiplicative scheme that treats zero exactly implies `P=NP`. No conclusion follows for a coarse additive error scaled by all `11^h` assignments, and no `BPP/BQP` collapse follows.

8 Limitations

The theorem is total on a natural, polynomially recognizable guarded family; it does not establish `#P` membership for unrestricted topological completion. It counts an exact recovered core or blank-padding equivalence class, not every padded rectangle. It does not require or imply a macrotile parser. Targets are unoriented classical knot types; framed, oriented, satellite, concordance, and multi-component target classifications are not claimed. Jones equality is used only on the proved split-union family, not as a general polynomial-time predicate.

9 Reproducibility

The raw baseline regressions exhaust all local source types and several finite graphs. The tagged regression replaces one forced edge of the marked `2x4` ladder with a literal protected trefoil tangle, recovers the two expected two-factors with component histogram `{1:1,2:1}`, verifies that tag erasure preserves both component counts, and checks determinant three on the tagged one-cycle example. Corruption tests alter the tag, gate, moat, and source marking and must be rejected.

The exact commands, environment, hashes, and evidence limits are recorded in `08_REPRODUCIBILITY.md`. Finite tests guard implementation details; Lemmas 3.1 and 4.1–4.3 and the tag-erasure argument prove the unbounded statements.

10 References

The complete bibliography, including foundational work and the 2021–2026 search record, is in `07_REFERENCES.bib`; the theorem-by-theorem inclusion and exclusion matrix is `06_LITERATURE_MATRIX.csv`.

Data and code availability

The accompanying repository contains claim registries, machine-readable witnesses, independent finite regressions, and a browser-native Pyodide demonstrator. The paper distinguishes finite computational checks from the symbolic arguments proving the all-size statements.

Declarations

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